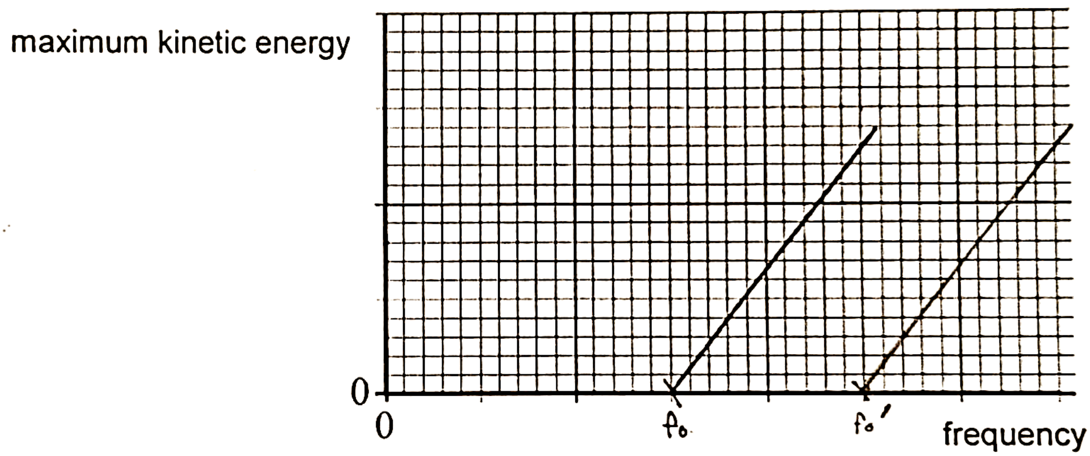
8 **(a)(i)** A photon is a discrete packet (quantum) of energy of electromagnetic radiation. **B1**

(ii) The work function energy of a metal is the minimum energy of photon to cause emission of electron from the surface of a metal. **B1**

(b)(i)



1. f_0 correctly labelled **B1**
2. parallel line with same gradient and higher horizontal intercept **B1**

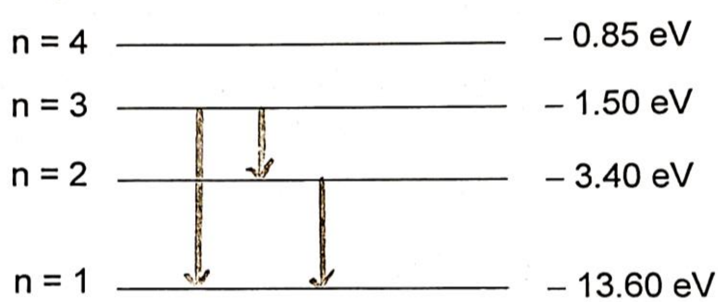
(ii) Planck constant **B1**

(iii) Max KE corresponds to electrons emitted from surface
Energy is required to bring electron to surface **B1**

(iv) intensity determines number of photons arriving per unit time **B1**
Hence determines number of electrons emitted per unit time,
and not energy. **B1**

(c)(i) 13.60 eV **B1**

(ii)1.



correct energy levels **B1**
correct direction of arrows **B1**

2. energy in Joules = $1.90 \times 1.6 \times 10^{-19} = 3.04 \times 10^{-19}$ **C1**

$E = hc/\lambda$

$\lambda = (6.63 \times 10^{-34})(3 \times 10^{8}) / (3.04 \times 10^{-19})$

$= 6.54 \times 10^{-7}$ m **A1**

Visible light region (red) **B1**

(d)(i) Wavelength that is associated with a particle that is moving. **B1**

(ii) $\lambda = h/mv$ **C1**

$v = (6.63 \times 10^{-34}) / (9.11 \times 10^{-31})(1.2 \times 10^{-10})$ **C1**

$= 6.1 \times 10^{6}$ m s^{-1} **A1**

(iii) wavelength is about the separation of atoms in a crystal **B1**

can be used in electron diffraction **B1**

9 **(a)(i)** The average time taken for the initial number of nuclei (or activity) of **B1**